Syllabus

Summer School in AI & Applied Economics Course Syllabus

Zurich, Summer 2026
(link to this page)

Organizers: Elliott Ash, Sergio Galletta, Joachim Voth, David Yanagizawa-Drott
Instructors: Andrea Ciccarone, Felix Drinkall, Germain Gauthier, Philine Widmer, Fabian Roeben
Advisors: Felix Chopra, Carlo Schwarz

Overview

Recent advances in artificial intelligence are reshaping both the economy and the empirical toolkit available to economists. Machine learning, natural language processing, computer vision, and generative AI are expanding the scope of measurable phenomena and enabling new research designs in applied economics. The Zurich Summer School in AI & Applied Economics is designed to train PhD students in applying these methods to frontier empirical research. The program focused on the full research pipeline: from framing economic questions, to constructing datasets with AI methods, to identification and inference.

Preparation and access

1. Participants should bring a laptop suitable for coding and should complete the setup steps before Week 1.
2. Arrange access to at least one capable AI service, such as ChatGPT Plus (or higher) or Claude Pro (or higher). Participants for whom this is not feasible should contact the organizers about course-provided access.
3. Install at least one coding agent: Codex and/or Claude Code.
4. Install Git and a working Python/Jupyter environment.

Week 1 – August 31st - Sept 4th, afternoons CET

on Zoom (https://ethz.zoom.us/j/64612801976)

Mon 13:00-14:00 Course introduction – Elliott Ash
Mon 14:30-16:00 ML basics + embeddings/clustering – Andrea Ciccarone
Mon 16:30-18:00 Text + image data – Andrea Ciccarone

Tue 14:00-15:45 ML and econometrics – Germain Gauthier
Tue 16:15-18:00 LLMs/alignment – Elliott Ash

Wed 14:00-15:45 Working with agents – Fabian Roeben
Wed 16:15-18:00 Software/app development with AI – Fabian Roeben

Thu 14:00-15:45 Training LLMs – Felix Drinkall
Thu 16:15-18:00 TBD (maybe nothing)

Fri 14:00-15:00   Recap/conclusion/presentations – Elliott Ash

Week 2 – Sept 7th-10th, all day

ETH Zurich RZ Building, Room F21

Day	Session	Time	Activity
MON 7 SEP	Morning	09:00–12:30	Sergio Galletta
	Afternoon	14:00–15:45	Sergio Galletta
		16:00–17:30	Project Studio
TUE 8 SEP	Morning	09:00–12:30	Elliott Ash
	Afternoon	14:00–15:45	Philine Widmer
		16:00–17:30	Project Studio
WED 9 SEP	Morning	09:00–12:30	Joachim Voth

	Afternoon	14:00–17:30	Project Studio and advisor meetings
THU 10 SEP	Morning	09:00–12:30	David Yanagizawa-Drott
	Afternoon	14:00–17:30	Project Studio, consultations and next steps

Instructor Sessions and Suggested Readings

Monday, 7 September — Sergio Galletta (09:00–12:30 and 14:00–15:45)

The session will connect machine learning for anti-corruption policy with bots as experimental treatments.

Suggested papers and materials:

- A Machine Learning Approach to Analyze and Support Anti-Corruption Policy
- BallotBot: Can AI Chatbots Lower Voter-Information Barriers?

Additional papers:

- Fiscal Policy Divergence (with Piera Bello and Costanza Marconi), using cosine similarity across municipal budgets
- The Partisan Divergence in U.S. Local Policy, 1972–2017 (with Elliott Ash), using predictive accuracy to measure partisan differences in local policy.

Tuesday, 8 September — Elliott Ash (09:00–12:30)

- Worker Rights in Collective Bargaining
- Courts of Tomorrow: Evidence from a Nationwide Rollout of Generative AI

Tuesday, 8 September — Philine Widmer (14:00–15:45)

- Chorale: Deep Latent Variable Models for Social Science.

Wednesday, 9 September — Joachim Voth (09:00–12:30)

- Revealing Life Preferences Through LLMs (April 2026)
- Giulia Caprini, Visual Bias.

Additional topics: Vision LLMs at Work — Image(s) and Lolitas, and Time-Locked LLMs.

Thursday, 10 September — David Yanagizawa-Drott (09:00–12:30)

- Automation with Generative AI? Evidence from a Teacher Hiring Pipeline
- Verifying the Verifiers: Towards Autonomous Policy Evaluation

Assignments

Week 1: problem set

Participants will complete a problem set based on the methods and workflows introduced during the online week. TBD: confirm the scope, release date, deadline, and submission method.

Week 2: research idea

Participants will develop a research idea that uses AI as a research method, an object of study, or a tool for building a research application. TBD: confirm the expected format, deadline, and whether ideas will be presented during the in-person week.

Bibliography

Books

- Grimmer, Roberts, and Stewart, Text as data
- Geron, Hands-on machine learning with scikit-learn
- Raschka, Build a large language model (from scratch)
- Jurafsky and Martin, Speech and language processing
- Goldberg, Neural networks for natural language processing
- Lambert, RLHF Book

Review Papers

- Gentzkow, Kelly, and Taddy, "Text as Data." JEL 2019.
- Ash and Hansen, "Text Algorithms in Economics." ARE 2023.
- Dell, Deep learning for economists. JEL 2025.
- Hassan et al, Text as data in economic analysis. JEP 2025.
- Ludwig et al, Large language models: An applied econometric approach. ARE 2026.

Articles

1. Adukia et al, What we teach about race and gender: Representation in images and text of children's books, QJE 2023.
2. Angelucci et al, Policymaking in the American States, 1787-2020.
3. Antoniak, Mimno, and Levy, Narrative Paths and Negotiation of Power in Birth Stories.
4. Ash, Chen, and Ornaghi, Gender attitudes in the judiciary: Evidence from U.S. Circuit Courts, AEJAE 2024.
5. Ash et al, BallotBot: Conversational AI and Voter Information Acquisition.
6. Ash and Xue, British Industrialization and Cultural Change: Evidence from Proverb Use.
7. Ash, Gauthier, and Widmer, Relatio: Text Semantics Capture Political and Economic Narratives, PA 2024.
8. Ash et al, "Worker Rights in Collective Bargaining".
9. Ash et al, "More Laws, More Growth? Evidence from U.S. States", JPE 2025.
10. Ash et al, "A Machine Learning Approach to Analyze and Support Anti-Corruption Policy", AEJEP 2025.
11. Awuah et al, Automation using Generative AI? Evidence from a Teacher Hiring Pipeline.
12. Bertrand et al, Hall of Mirrors: Corporate Philanthropy and Strategic Advocacy, QJE 2021.
13. Bloom et al, Remote work across jobs, companies, and space
14. Braghieri et al, Article-Level Slant and Polarization of News Consumption on Social Media
15. Caprini, Visual Bias
16. Chopra and Haaland, Conducting qualitative interviews with AI
17. Chopra et al, News Customization with AI.
18. Ciccarone, An image is worth 1000 words? Partisanship and attention in videos
19. Gennaro and Ash, Emotion and Reason in Political Language, EJ 2022.
20. Gottlich et al, History LLMs: Giving the Past a Voice with Large Language Models
21. Haaland et al, Understanding economic behavior using open-ended survey data.
22. Habibi et al, The content moderator's dilemma. EJ 2026.
23. Haq et al, Revealing Life Preferences through LLMs

24. Hansen, McMahon, and Prat, Transparency and deliberation with the FOMC: A computational linguistics approach, QJE 2017.
25. Hassan, Hollander, Van Lent, and Tahoun, Firm-Level Political Risk: Measurement and Effects, QJE 2019.
26. Lagakos et al, American Life Histories
27. Longuet-Marx, Party lines or voter preferences? Explaining political realignment
28. Ludwig and Mullainathan. "Machine Learning as a Tool for Hypothesis Generation." QJE 2024.
29. Moreno-Medina et al, Officer-Involved: The Media Language of Police Killings. QJE 2025.
30. Mehmood, Goessmann, Ash, Courts of Tomorrow: Evidence from a Nationwide Rollout of Generative AI.
31. Voth and Yanagizawa-Drott, Image(s).
32. Widmer, Abed-Meraim, Galletta, and Ash, Media slant is contagious, EJ 2026.
33. Willner and Yanagizawa-Drott, Verifying the Verifiers: Towards Autonomous Policy Evaluation.

Methods Material

- Angelopoulos, Bates, Fannjiang, Jordan, and Zrnic (2023). "Prediction-Powered Inference." Science 382(6671): 669-674.
- Battaglia, Christensen, Hansen, and Sacher (2024). "Inference for Regression with Variables Generated by AI or Machine Learning." arXiv:2402.15585.
- Egami, Hinck, Stewart, and Wei (2023). "Using Imperfect Surrogates for Downstream Inference: Design-Based Supervised Learning for Social Science Applications of Large Language Models." Advances in Neural Information Processing Systems 36 (NeurIPS).
- Denny and Spirling, "Text Preprocessing for Unsupervised Learning: Why It Matters, When It Misleads, and What to Do about It."
- Kohler et al, "Read the Paper, Write the Code: Agentic Reproduction of Social-Science Results"
- Antoniak, Topic modeling for the people
- Spirling and Rodriguez, Word embeddings: What works, what doesn't, and how to tell the difference for applied research.
- Yoav Goldberg and Omer Levy, "Word2Vec explained: Deriving Mikolov et al's Negative Sampling Word Embedding Method".
- Bloem, Transformers from scratch
- Huggingface Transformers Summary of Models
- Nathan Lambert, DeepSeek R1's recipe to replicate o1 and the future of reasoning LMs
- Nicolas Bustamente, The RAG Obituary: Killed by Agents, Buried by Context Windows
- Törnberg (2024). "Best Practices for Text Annotation with Large Language Models."